

Subalgebras and Filters; Stone's Representation Theorem

Before we address the representation theorem in the general case, let us define a couple more of basic notions in Boolean algebras.

Consider a Boolean algebra $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$.

Definition 0.1. (Subalgebra)

A *subalgebra* $\mathcal{C} := (C, \leq, \vee, \wedge, ^c, 0, 1)$ of $\mathcal{B}$ consists of a subset C of B such that C forms a Boolean algebra with respect to the same operations and least and top elements as in $\mathcal{B}$.

Clearly, $(\{0, 1\}, \leq, \vee, \wedge, ^c, 0, 1)$ and $\mathcal{B}$ itself, are trivial subalgebras of $\mathcal{B}$.

An example of a Boolean subalgebra of a power set Boolean algebra $(\mathcal{P}(U), \cup, \cap, ^c, \emptyset, U)$ with U infinite, is the one formed by the set of all finite and *cofinite* subsets (i.e. those whose complements are finite) of U .

Definition 0.2. (Filter) Let F be a subset of B . F is a *filter*, if it satisfies the following properties.

- (i) $F \neq \emptyset$.
- (ii) If $a, b \in F$ then $a \wedge b \in F$.
- (iii) If $a \in F, a \leq b$ then $b \in F$.

B is trivially a filter of itself.

So we define F to be a *proper filter*, if it is a proper subset of B or equivalently, if 0 does not belong to F .

F is a *prime filter* if it is proper and if $a \vee b \in F$ implies that either a or b belongs to F .

By $F(a)$, we denote the smallest filter in $\mathcal{B}$ that contains a , $a \in B$. It is called the *principal filter generated by a* .

It can be shown that $F(a) = \{x \in B : a \leq x\}$. If $a \neq 0$, $F(a)$ is a proper filter. Thus every element of a Boolean algebra gives rise to a (proper) filter.

Observe also that if F is a prime filter, then for any element $a \in B$, exactly one of a and a^c is in F .

We are now ready to present the Stone's representation theorem – we shall only give the main ideas of the proof. Consider the Boolean algebra $\mathbf{2} := (\{0, 1\}, \leq, \vee, \wedge, \neg, 0, 1)$, and let $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$ be any Boolean algebra.

Two fundamental results about filters in a Boolean algebra will be assumed at the moment:

- (a) There is a prime filter containing any proper filter. In particular, there is a prime filter containing $F(a)$, for any $a(\neq 0) \in B$.
- (b) For any prime filter F , there is a map $h : B \longrightarrow \{0, 1\}$ such that h is a homomorphism from $\mathcal{B}$ to $\mathbf{2}$, and $F = \{x \in B : h(x) = 1\}$.

We shall, in fact, give the proof of (a) later, while discussing *Zorn's Lemma*. The result in (b) is left as an exercise.

Theorem 0.1. (Stone's Representation Theorem)

$\mathcal{B}$ is isomorphic to a subalgebra of the power set Boolean algebra of some set.

Proof. Let $\mathcal{H}(\mathcal{B})$ denote the set of all homomorphisms from $\mathcal{B}$ to $\mathbf{2}$.

Define a map $H : B \longrightarrow \mathcal{P}(\mathcal{H}(\mathcal{B}))$ as follows.

For any $b \in B$,

$$H(b) := \{h \in \mathcal{H}(\mathcal{B}) : h(b) = 1\}.$$

Claim: For any $a, b \in B$, $a \leq b$ if and only if $H(a) \subseteq H(b)$.

That H is one-one would follow from the claim, and using earlier proposition, it would also be established that H is an isomorphism between the Boolean algebras $\mathcal{B}$ and $H(\mathcal{B})$. Note that $H(\mathcal{B})$ is a Boolean algebra defined over the image set $H(B) := \{H(a) : a \in B\}$ of B under the map H , and is a subalgebra of the power set algebra of $\mathcal{H}(\mathcal{B})$.

Proof of Claim: Let $a \leq b$ and $h \in H(a)$. Then $1 = h(a) \leq h(b)$, and so $h \in H(b)$.

Now suppose $a \not\leq b$. This means $a \wedge b^c \neq 0$. Consider the filter $F(a \wedge b^c)$ – it is then a proper filter. Thus by result (a) assumed above, there is a prime filter F_0 containing $F(a \wedge b^c)$. As $a \wedge b^c \in F_0$, both $a, b^c \in F_0$.

By result (b) assumed above, for F_0 there is a homomorphism h_0 from $\mathcal{B}$ to $\mathbf{2}$ such that $F_0 = \{x \in B : h_0(x) = 1\}$. Finally, as $a, b^c \in F_0$, we have $h_0(a) = 1$, while $h_0(b) = 0$. Thus $h_0 \in H(a)$, but $h_0 \notin H(b)$. $\square$